

Complete solutions to all problems are required. Models should be specified, approximations and conclusions should be specified and motivated.

Problems 1, 2 and 3 can give a maximum of 10 points each. Problems 4, 5 and 6 can give a maximum of 20 points each. The maximum total on the exam is 90 points. The limit for Pass (P) is at least 45 points, and the limit for High Pass (HP) is at least 67 points.

Only the paper sheets provided by the institute in the exam room shall be used, both for solutions and for sketches/notes.

Each problem solution shall start at the top of a new paper.

Write your ID number on each paper that you hand in.

Red pencil or pen is not allowed.

Calculators are allowed. Only statistical tables provided by the institute are allowed. No other tables or formula sheets are allowed.

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1. Suppose that T is a random variable taking values in $[0, \infty)$. Let $\theta = E(\sin(T) \cdot 1\{0 \leq T < 2\pi\})$ be a parameter of interest. The following are i.i.d. observations of T

6.39, 2.63, 0.572, 2.51, 7.43, 23.7, 3.39, 10.8, 7.57, 42.0, 2.77, 18.9, 64.7, 6.42, 6.90, 16.8.

Find the plug-in estimator of θ . [10p]

2. The time T until a radioactive particle decays is assumed to be distributed according to the exponential distribution, with the expected time to decay, θ , assumed to be unknown. In an experiment when one isolates one particle and observes it until it decays, one observes that it decays after 1450 minutes.

(a) Find the maximum likelihood estimator of θ . [5p]

(b) Construct an upper bounded 95% confidence interval for θ . [5p]

3. In an experiment one performs repeated independent trials until a trial succeeds for the first time, where in each trial the probability of a success is θ . The experiment is done three (independent) times, and the results are that one needs 5, 8 and 2 trials, respectively, until one succeeds for the first time, in the three experiments.

(a) Find the maximum likelihood estimator of θ . [5p]

(b) Make an exact test of the hypotheses

$$H_0 : p \geq 1/3,$$

$$H_1 : p < 1/3.$$

[5p]

4. When measuring the acidity level in the ocean outside of a large city the measurements may be considered to be Gaussian. One may assume that the expected acidity level may vary between days but that the variance, θ , in the acidity on different days stays the same. One makes several measurements (during one hour) of the levels, on two consecutive days. The following are the (two sets of independent) i.i.d. observations of the acidity levels on the two days:

Day 1 6.38, 4.99, 5.98, 4.77, 6.53, 6.44, 3.75,
 Day 2 9.30, 8.74, 9.20, 9.75, 9.80, 9.61, 8.88, 8.93.

- (a) Derive the maximum likelihood estimator of θ . [10p]
 (b) Construct a two sided 95% confidence interval for $\log(\theta)$. [10p]
5. A medical research facility wants to explore if the maximal oxygen uptake (VO2-max) for heart patients (all in their 50's) can be improved with exercise and diet. One models VO2-max y , as measured for an individual on a cycle ergometer, as a function of the amount x_1 (minutes/week) of running, as well as the amount x_2 (mg/day) of consumed theobromine (active substance in dark chocolate) and the amount x_3 (μ g/day) of vitamine D (given as a supplement). The hospital wants to see if and with how much exercise and diet increase fitness. The model proposed for analyzing the data is

$$y_i = \theta_0 + \theta_1(x_{1i} - \bar{x}_{1.}) + \theta_2(x_{2i} - \bar{x}_{2.}) + \theta_3(x_{3i} - \bar{x}_{3.}) + \epsilon_i,$$

for $i = 1, \dots, n$, with ϵ_i i.i.d. $N(0, \sigma^2)$ -distributed random variables, and $\sigma^2 = 4$ assumed to be known.

The hospital examines 30 individuals in a study. The summarised data are, on matrix form,

$$(X^T X)^{-1} = \begin{pmatrix} 3.33 \cdot 10^{-2} & 0 & 0 & 0 \\ 0 & 8.16 \cdot 10^{-5} & -4.92 \cdot 10^{-5} & -3.21 \cdot 10^{-4} \\ 0 & -4.92 \cdot 10^{-5} & 3.16 \cdot 10^{-4} & 1.37 \cdot 10^{-4} \\ 0 & -3.21 \cdot 10^{-4} & 1.37 \cdot 10^{-4} & 1.39 \cdot 10^{-2} \end{pmatrix},$$

$$X^T Y = \begin{pmatrix} 1204.4 \\ 4589.0 \\ 936.18 \\ 94.856 \end{pmatrix},$$

$$\bar{x}_{1.} = 60.7,$$

$$\bar{x}_{2.} = 20.2,$$

$$\bar{x}_{3.} = 16.5.$$

- (a) Does exercise and/or diet increase fitness (measured as VO2-max)? (You do not have to take into account multiple testing problems.) [6p]
 (b) How much does an increase of 10 minute/week more exercise increase the fitness, when the diet supplements are kept fixed? [6p]
 (c) Construct a 95 % confidence interval for VO2-max for an individual that exercises 100 min/week, eats 70 mg theobromine and 10 μ g vitamine D per day. [8p]
6. The continuous random variable V has an unknown probability density function F . One obtains the following observations of V

4.93, 5.67, 5.43, 4.76, 5.57, 5.23, 4.36, 5.60, 5.47, 5.98.

- (a) Find the plug-in estimator of the upper 20 percent quantile $q_{0.2}$ of F . [5p]
(b) Make an exact test of the hypotheses

$$H_0 : \quad q_{0.2} \leq 5.5,$$

$$H_1 : \quad q_{0.2} > 5.5.$$

- [5p]
(c) Suppose that one has 75 observations of V and that the estimated value of $q_{0.2}$ has the same numerical value as in *a*). Construct an approximate test for the above hypotheses, at significance level 0.05, and calculate the (approximate) power of that test at $q_{0.2} = 5.7$. [10p]